

Albert Xu

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EDUCATION

University of California, San Diego

Sep 2022 - Mar 2024

M.S in Computer Science (GPA: 3.88/4.0)

La Jolla, CA

- Courses: Principles of Computer Operating Systems, Principles/Program Languages, Statistical NLP, Principles of Computer Architecture, Computer Science and Engineering, Principles of Machine Learning, Principles of AI

University of Liverpool

Sep 2018 - Jun 2022

B.S in Computer Science (GPA: 3.93/4.0, top 3%)

Liverpool, UK

TECHNICAL SKILLS

Programming Languages: Python, Java, Go, SQL, C/C++, HTML/CSS, JavaScript, Rust, Objective-C, C#, R, Haskell

Frameworks and Tools: Django, React, Git, Docker, AWS, gRPC, Matlab, Flask, Hadoop, DynamoDB

EXPERIENCE

4Pexonic Inc.

Oct 2024 - Present

Back-end Engineer

San Diego, CA

- Developed an academic impact **analysis platform** tracking citation patterns and research influence using Node.js, MongoDB, and Next.js with **Docker** containerization, serving university researchers and policy institutions
- Built two-tier caching with in-process **LRU** and **Redis** for bibliometric queries, reducing response times by **40%** for complex research trend analysis and cross-disciplinary impact calculations
- Integrated **gRPC** microservices for data processing pipelines and implemented secure authentication with **bcrypt**/session management, configured **Nginx SSL** to protect academic publication data and researcher information
- Created API documentation for citation tracking endpoints, established monitoring for data accuracy validation, and collaborated with QA team on bibliometric testing protocols, improving system reliability by **25%**

UCSD ECE Department

Jan 2024 - Apr 2024

Web Developer

La Jolla, CA

- Developed the ITA Workshop [website](#) with video/document upload functionality using **Django** and **React**, serving 1000+ users with Redis server-side caching and optimized SQL indexes for improved page load performance
- Refactored file upload system to **asynchronous** chunked processing, enabling large video and document handling while maintaining ACID-safe database schema changes and improving overall system throughput
- Implemented multipart S3 uploads with SHA-256 integrity checks and **PostgreSQL** performance tuning, achieving high-throughput file transfers with cross-region replication for data durability

UCSD Halcioğlu Data Science Institute

Oct 2023 - Sep 2024

Research Engineer

La Jolla, CA

- Built a modular “Injector” pipeline (pattern-gen → sampling → injection → evaluation) with **beam-search** pruning and **Optuna** Bayesian tuning, raising downstream model accuracy by **10%** and fairness by **15%**
- Co-authored “[Stress-Testing ML Pipelines with Adversarial Data Corruption](#),” VLDB 2025; introduced the first adversarial benchmark for data-quality robustness
- Ran comprehensive **Inject** → **Clean** → **Retrain** benchmarks on missing-value, selection-bias, and outlier scenarios while optimizing Z-score standardization for skewed data—boosting attack coverage **30%**

PROJECTS

SurfStore: Distributed Cloud Storage Server Project | [Link](#)

- Designed and implemented a **Dropbox**-like cloud-based file storage service using **Go**, featuring distributed storage and consistent hashing algorithms for efficient file management
- Built for large-scale data handling, the project showcased excellent scalability during tests and efficiently met the data storage demands of thousands of users through client-server communication via **gRPC**
- Enhanced the system's fault tolerance by implementing the **RAFT** distributed consensus protocol, which led to a **50%** improvement in error recovery efficiency during testing